

problem of what triggers an economic crisis and the problem of the inevitability of economic crisis mutually relate to each other, they are different issues.

4.3.1 Inevitability of Economic Crisis

4.3.1.1 Meaning of Inevitability

If one wants to prove the inevitability of economic crisis in the capitalist system, he only has to show that it is inevitable as long as the capitalist system exists. In order to do that, one only has to prove the following two propositions: (1) In the capitalist system, it is inevitable that a disequilibrium cumulates; (2) If the disequilibrium cumulates unlimitedly, it will be impossible for the capitalist system to survive or reproduce itself.

If those two propositions are proven, it follows from combining them that it is absolutely necessary to *reverse* the cumulative process of disequilibrium in order for the capitalist system to survive or reproduce itself. Since the cumulative disequilibrium is unavoidable in the capitalist system and the capitalist system would not last without the reversal, whatever the trigger, the cumulative process of disequilibrium must be reversed as long as we premise the continued existence and reproduction of the capitalist system. The economic crisis is the event that the *upward* cumulative process of disequilibrium is reversed and switched to a *downward* cumulative process of disequilibrium.

Let us examine the above two propositions in sequence to prove the inevitability of economic crisis.

4.3.1.2 Inevitability of Cumulative Disequilibrium

As mentioned in Chap. 2, a capitalist society is a class society in which the production of commodities is dominant. Whereas production is made by the social division of labor, decisions about each part are made privately by private individuals who own the means of production privately, and there is no social decision or planning that brings together the whole social production. Consequently, each private producer has to make decisions on production capacity or direction/level of production without knowing social demand in advance. Therefore, it is inevitable for a disequilibrium, such as a gap between actual production and production capacity or the difference between production and social demand, to arise.

However, even though a disequilibrium arises, it does not necessarily mean that the disequilibrium cumulates or that it hinders the reproduction of society. For instance, when the commodity production exists as a non-dominant mode so that the commodity exchange is not necessary as a medium for production goods replacement or labor reproduction, which are essential factors of reproduction, a disequilibrium might arise in the commodity production, but it does not hinder the reproduction of

society. Besides, even if the commodity production exists broadly in society, when the sole commodity production is implemented with one's own labor using one's private means of production without wage laborers, private commodity producers' characteristic incentive is not profit pursuing but use-value acquisition through exchanges. As a result, the gap between production and consumption for a whole society is small, for which a partial disequilibrium might arise, but a general disequilibrium is unlikely to happen.

In the capitalist system, not only is commodity production dominant, but also labor becomes a commodity. Capitalist commodity production with two classes—wage laborers and capitalists—is implemented. The capitalists make production decisions, and their characteristic incentive is profit pursuit. Consequently, exploitation of wage laborers by capitalists prevails. When there is exploitation, inequality always holds for society as a whole.

$$X > C + V \quad (4.19)$$

Here, X is the value of aggregate production, C is the value of production goods exhausted to produce X , and V is value of consumption goods paid as wages to the laborers employed to produce X . The inequality (4.19) shows that there exists exploitation. In fact, (4.19) can be rewritten as the following formula.

$$X - C - V = N - V = S > 0 \quad (4.20)$$

Here, N is total labor in the current period that the laborers sold and S is surplus labor. The inequality (4.19) shows that the aggregate production cannot be realized at a price inclusive of profits only with the demand that replaces production goods exhausted in the current period and the consumption demand of the laborers. In fact, in the right-hand side of (4.19) C shows replacement demand for production goods and V consumption demand of the laborers, but the total of those two is smaller than X ; value of aggregate production.

That is why there must be some other demand besides replacement demand for production goods and consumption demand of the laborers to realize commodities smoothly in a capitalist system. Since the wage labor class is supposed to spend their entire wage, this additional demand must come from the capitalists. The capitalists make consumption goods demand for private consumption. Let us denote it S_k in terms of value. The demand is short even after this private consumption of the capitalists is added so that we have the following inequality:

$$X > C + V + S_k \quad (4.21)$$

It is because capitalists do not exploit the laborers only for their private consumption. If so, the exploitation is bounded by their limit of private consumption, that is, the size of their appetite. Unlike the ruling class in the class society before the capitalist system, the capitalist class exploits not only for their own, private consumption, but also for greater accumulation. Therefore, in a capitalist system, it holds always as follows:

$$S > S_k \quad (4.22)$$

Meaning, surplus value outweighs the private consumption of capitalists. If (4.22) holds, (4.21) always follows because of (4.20).

Equation (4.21) shows that the aggregate production for a whole society outweighs the sum of replacement for production goods, and the private consumption of laborers and capitalists. It means that it shows the “contradiction between production and consumption”. It can be understood from the explanation above that the contradiction between production and consumption arises from (1) that laborers are exploited, and (2) that the exploitation is made by capitalist not only for their consumption but also for accumulation.

Now, despite the contradiction between production and consumption shown by (4.21), there must be additional demand in order for aggregate production to be realized: capitalists’ accumulation demand S_a . Capitalists try to increase production capacity in the next period and after by demanding for more production goods than what is exhausted for production in the current period. The question here is the size of accumulation demand S_a . If the balance that deducts capitalists’ private consumption from surplus value is always equal to the accumulation demand, that is

$$S_a \equiv S - S_k \quad (4.23)$$

from (4.23) and (4.20),

$$X \equiv C + V + S_k + S_a \quad (4.24)$$

the aggregate production and the aggregate demand for a whole society always match. Therefore, even if there are disequilibria in individual sectors, those disequilibria offset each other for the whole society, and no general disequilibrium arises.

However, in a capitalist system, (4.23) is *not always* guaranteed to be true. As repeatedly emphasized, decisions on production level and production capacity are made by private capitalists. Private capitalists have no choice but make decisions on accumulation demand based on their private expectations, without general information or control instruments on social production. Therefore, when his expectations on the market conditions are not promising, a capitalist may not implement accumulation demand in the amount that is equal to the residual of the gained profits after subtracting private consumption. When his expectation is bullish, he may increase accumulation demand with credit beyond the limitation of his gained profits.

What is more problematic is that, as explained in detail in Chap. 3, capitalists do not implement accumulation demand after profits are realized. How much accumulation demand they make as a whole determines the size of realized profits. Therefore, for most capitalists, they do not implement accumulation demand with realized profits; they have no choice but to expect profits and determine accumulation demand before profits are realized. For that reason, (4.23) does not hold in a capitalist system so that the aggregate production and the aggregate demand do not match and that a general disequilibrium arises.

Even if the general disequilibrium arises, the cumulative movement of disequilibrium will not occur when the general disequilibrium is cleared by the price adjustment mechanism. When the general disequilibrium arises in the commodity markets, for instance,

$$X < C + V + S_k + S_a \quad (4.25)$$

is the case, someone may think that the equilibrium can be recovered smoothly with a production increase and demand contraction induced by a price increase. He/she commits the mistake of applying the price adjustment mechanism in the single commodity market to the whole society. As examined in Sect. 4.1, whether the general disequilibrium moves toward equilibrium or away from it depends on the movement of accumulation demand. For example, when there is disequilibrium as (4.25), if accumulation demand S_a decreases, it converges to the equilibrium; on the contrary, if it increases, the disequilibrium strengthens. When facing a general disequilibrium, it is the personal decisions of private capitalists which determines the adjustment direction of the accumulation demand. Private capitalists do not make decisions based on what kind of adjustment of accumulation demand is needed in order to recover the equilibrium of the whole society. They try to recover the private equilibrium from their own private point of view. Most capitalists have problems such as excess demand or shortage of production capacity when (4.25) is the case. In this instance, to recover their private equilibrium, increasing accumulation demand becomes urgent. Thus, the disequilibrium cumulates in the society as a whole.

As you can understand from the above explanation, in a capitalist system, (1) although the production is implemented in the system of social division of labor, its individual components depend on private decisions, (2) capitalists exploit wage laborers, (3) capitalists exploit not only for their private consumption, but also for the accumulation, and (4) private capitalists make decisions on accumulation privately and dispersedly. As a result, a general disequilibrium arises, and the disequilibrium must be cumulative.

4.3.1.3 Violation of Capitalist Reproduction

An economy is in the upward cumulative disequilibrium process when a general disequilibrium exists and cumulates as a whole economy, in the way that aggregate demand exceeds aggregate supply or that the actual amount of production exceeds normal production capacity. The issue here is whether capitalistic production relations can be sustained *if* this upward cumulative disequilibrium process continues unlimitedly. It should be noted in advance that the author does *not* claim here that the upward cumulative disequilibrium process *does* continue unlimitedly. Our goal is to present that the cumulative disequilibrium process cannot continue without limit as long as a capitalist system is sustained, by showing that the capitalist system would not be able to survive if it continued unlimitedly.

Capitalists regard the upward cumulative disequilibrium process not as disequilibrium, but as a continuous favorable situation. In fact, no matter how much production they make, products fly off the shelves, and the profit rate is high. They cannot stop laughing, literally. Yet, regardless of whether it is favorable or not for capitalists, this is the cumulative disequilibrium process. As already examined, the main characteristic of this process is that accumulation demand is increasing acceleratively. From another point of view, it is the process in which the gap between production and consumption is increasing fast. In fact, it is the process in which the larger part of the aggregate production is absorbed for the enhancement of production capacity and the smaller part that is left is devoted to private consumption. This can be found in the increasing exploitation rate of the laborers in this process. The real wage rate of the laborers must increase slower than labor productivity growth. It even absolutely decreases when the increase rate of accumulation demand is exceptionally high, even if there is an increase in labor productivity.

These situations are not explicit at the beginning of the upward cumulative disequilibrium process. It is because that even if accumulation demand increases at an accelerated rate, the aggregate production level can be increased relatively easily when there are huge idle production capacity and unemployment. As long as the increase in accumulation demand is absorbed by the increase in production level, the gap between production and consumption increases *absolutely*, but it does not necessarily increase *relatively*. That is, private consumption can also increase as accumulation demand increases. Moreover, in that case, an increase in the exploitation rate is not inevitable. If there is an increase in the exploitation rate or a fall of the real wage rate, the real wage in terms of consumption goods that the laborers obtain for labor per day will increase as the result of the rise in employment or labor hours per day.

However, when idle production capacity is not left unutilized, and when unemployment is absorbed to the extent in which it becomes difficult to enhance production levels to satisfy increasing accumulation demand by extending operation, the contradiction between production and consumption becomes clear in an intense form. Namely, at this stage of the process, it becomes unavoidable to squeeze private consumption in order to cover increased accumulation demand. The gap between production and consumption has to increase not only absolutely, but also relatively. This relative compression of private consumption is realized through an increase in prices of consumption goods and the compression of real wage rates. A slowdown of the increase in real wage rates compared to labor productivity prevails clearly and, furthermore, it causes even absolute levels of real wage rates to fall.

Thus, the cumulative disequilibrium process proceeds while compressing private consumption both relatively and absolutely. What does it mean to the reproduction of capitalist production relations? It is not our subject now that the gap between production and consumption widens through the compression of private consumption, and it may cause a difficulty of realization in the market. As we will examine in Sect. 4.3.2, it is true that the gap between production and consumption causes overproduction in the consumption goods sector and that this can be a *momentum* for the turnover. However, this is not our focus right now, as we hope to reveal what kind of difficulty

it may cause to the sustainability of the capitalist reproduction if we assume that the accelerating increase in accumulation demand proceeds unlimitedly while enlarging the gap between production and consumption at an accelerating rate.

If the upward cumulative process of disequilibrium continues without limit, the capitalist production relations become impossible to reproduce. As the gap between production and consumption enlarges relatively and the increase in aggregate production by extending the operation slows down, private consumption begins to decrease and real daily wages of the laborers also have to decrease. Now, as we argued in Sect. 2.4, there are upper bounds and lower bounds that real wages must satisfy in order for capitalist production relations to be reproduced. To sustain capitalist production relations, wage laborers must appear regularly in the labor market for the sale of labor. For this to happen, real wages must not drop below a certain level. In fact, when real wages drop below the lower limit at which the wage laborers can barely meet their physiological necessity or falls below the income that can be earned by tiny enterprises that can be managed with little capital, the merit of being employed is lost for wage laborers. When real wages fall below certain lower bounds in these ways, wage laborers cease to appear in the labor market, and the foundation of capitalist production relations will stumble. Violent enforcement can make wage laborers work for wages for a while, but it cannot last forever.

Additionally, when the upward disequilibrium process continues unlimitedly, it becomes challenging for capitalist production relations to reproduce itself due to the following reasons as well: As the upward cumulative disequilibrium process continues, soon or later, unemployment shrinks and full employment will be reached. As long as the upward cumulative disequilibrium process continues, full employment keeps going on. Now then, what does the continuance of full employment in the labor market mean to the capitalist system? Supply and demand become tight in the labor market, and consequently, the laborers' bargaining power increases and the money wage rate increases sharply. However, if this is the only case, as long as accumulation demand increases at an accelerated rate, an increase in the price higher than an increase in the money wage rate is possible, which drives down the real wage rate, i.e., an increase in the exploitation rate can be maintained, despite the increase in the money wage rate.

Continuance of full employment undermines the foundation of capitalist production relations through other factors. In order for capitalist production relations to continue, it has to be possible for the capitalist class to purchase the labor power and make wage laborers work under capitalists' private decisions. That is, it has to be possible for them to make wage laborers work under the "discipline" of capital. However, as full employment is ordinary, it becomes difficult. The capitalist class has outstanding power over the wage labor class, and they are able to force them to work under their discipline because capitalists can fire laborers and consequently, they can take wage laborers' means of living away, i.e., they have the "power of life or death". If the situation in which capitalists are not able to dismiss laborers continues, sooner or later capitalists will not be able to make laborers work under their decisions. As it becomes impossible for capitalists to enforce laborers to obey their discipline, the capitalist class will no longer be able to keep on making decisions on production. It